

DBMS – Week 2

Project Title: Art Hub – An Online Art Marketplace & Supplies Store

PURPOSE: Identification of Entities, Attributes, Primary Keys, and Relationships

1. Entities and Attributes

➤ Customer

Primary Key: Customer _ID

Attributes:

- Customer _ID (PK)
- First _Name
- Last _Name
- Email
- Phone _Number
- Password
- Address
- City
- State
- Postal _Code
- Registration _Date

➤ Artist

Primary Key: Artist _ID

Attributes:

- Artist _ID (PK)
- Artist _Name
- Email
- Phone _Number
- Biography
- Portfolio _Link
- Profile _Image
- Join _Date

➤ Category

Primary Key: Category _ID

Attributes:

- Category _ID (PK)
- Category _Name

- Category _Type

➤ Artwork

Primary Key: Artwork _ID

Foreign Keys:

- Artist _ID
- Category _ID

Attributes:

- Artwork _ID (PK)
- Artist _ID (FK)
- Category _ID (FK)
- Artwork _Title
- Medium
- Description
- Price
- Dimensions
- Stock
- Image _URL
- Availability _Status

➤ Art Print

Primary Key: Print _ID

Foreign Key: Artwork _ID

Attributes:

- Print _ID (PK)
- Artwork _ID (FK)
- Print _Size
- Print _Type
- Price
- Stock

➤ 6. Art Supplies

Primary Key: Supply _ID

Foreign Key: Category _ID

Attributes:

- Supply _ID (PK)
- Category _ID (FK)

- Product _Name
- Brand
- Description
- Price
- Stock
- Image _URL

➤ Shopping Cart

Primary Key: Cart _ID

Foreign Key: Customer _ID

Attributes:

- Cart _ID (PK)
- Customer _ID (FK)
- Product _Type
- Product _ID
- Quantity

➤ Orders

Primary Key: Order _ID

Foreign Key: Customer _ID

Attributes:

- Order _ID (PK)
- Customer _ID (FK)
- Order _Date
- Total _Amount
- Order _Status
- Shipping _Address

➤ Order Details

Primary Key: Order Detail _ID

Foreign Key: Order _ID

Attributes:

- Order Detail _ID (PK)
- Order _ID (FK)
- Product _Type
- Product _ID
- Quantity

➤ Payment

Primary Key: Payment_ID

Foreign Key: Order_ID

Attributes:

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Method
- Payment_Date
- Amount
- Payment_Status
- Transaction_ID

➤ Delivery

Primary Key: Delivery_ID

Foreign Key: Order_ID

Attributes:

- Delivery_ID (PK)
- Order_ID (FK)
- Courier_Name
- Tracking_Number
- Dispatch_Date
- Delivery_Date
- Delivery_Status

➤ Review

Primary Key: Review_ID

Foreign Keys:

- Customer_ID
- Artwork_ID

Attributes:

- Review_ID (PK)
- Customer_ID (FK)
- Artwork_ID (FK)
- Rating
- Review_Text
- Review_Date

➤ Admin

Primary Key: Admin_ ID

Attributes:

- Admin_ ID (PK)
- Admin_ Name
- Email
- Password
- Role

2. Entity Relationship Diagram (Power BI Model View):

The following figure represents the Entity Relationship Model of the **Art Hub – An Online Art Marketplace & Supplies Store** designed using Power BI Model View. It illustrates the relationships between the entities based on their Primary Keys (PK) and Foreign Keys (FK), ensuring data integrity and efficient database management.



2. Relationships Between Entities

Relationship	Cardinality	Description
Customer → Orders	One-to-Many (1:M)	One customer can place many orders, but each order belongs to one customer.
Customer → Shopping Cart	One-to-One (1:1)	Each customer has one shopping cart, and each shopping cart belongs to one customer.
Customer → Review	One-to-Many (1:M)	A customer can write multiple reviews, but each review belongs to one customer.
Artist → Artwork	One-to-Many (1:M)	One artist can upload many artworks, while each artwork belongs to one artist.
Category → Artwork	One-to-Many (1:M)	One category can contain multiple artworks.
Category → Art Supplies	One-to-Many (1:M)	One category can contain multiple art supplies.
Artwork → Art Print	One-to-Many (1:M)	One artwork can have multiple print versions (A4, A3, Canvas Print, etc.).
Artwork → Review	One-to-Many (1:M)	One artwork can receive many customer reviews.
Orders → Order Details	One-to-Many (1:M)	One order contains multiple ordered items.
Orders → Payment	One-to-One (1:1)	Every order has one payment record.
Orders → Delivery	One-to-One (1:1)	Every order has one delivery record.

4. Summary of Keys

Primary Keys (PK)

- Customer_ID
- Artist_ID
- Category_ID
- Artwork_ID
- Print_ID
- Supply_ID
- Cart_ID
- Order_ID
- Order Detail_ID

- Payment_ID
- Delivery_ID
- Review_ID
- Admin_ID

Foreign Keys (FK)

- Artist_ID → Artwork
- Category_ID → Artwork
- Category_ID → Art Supplies
- Artwork_ID → Art Print
- Customer_ID → Shopping Cart
- Customer_ID → Orders
- Order_ID → Order Details
- Order_ID → Payment
- Order_ID → Delivery
- Customer_ID → Review
- Artwork_ID → Review

The **Art Hub – An Online Art Marketplace & Supplies Store** database consists of **13 entities** that efficiently manage customers, artists, artworks, art printouts, art supplies, shopping carts, orders, payments, deliveries, reviews, and administrators. The relationships between these entities ensure data consistency, eliminate redundancy, and support efficient order processing and inventory management. This relational database design provides a scalable and well-structured foundation for the successful implementation of the Art Hub platform.